

# US DOMESTIC FLIGHT DELAY PREDICTION

## Big Data Analytics & MLOps using Hive/Spark

|                                      |                                    |                                  |                                  |
|--------------------------------------|------------------------------------|----------------------------------|----------------------------------|
| <b>2,913,802</b><br>Operated Flights | <b>2019–2023</b><br>Five-Year Span | <b>3 ML Models</b><br>Algorithms | <b>AUC = 0.686</b><br>Best Model |
|--------------------------------------|------------------------------------|----------------------------------|----------------------------------|

### EXECUTIVE SUMMARY

#### *PROJECT OVERVIEW*

|                                                    |                                                      |                                                      |                                                       |
|----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|
| Total Flights Analyzed<br><b>2.9M</b><br>2019–2023 | Best Model AUC-ROC<br><b>0.686</b><br>GBT Classifier | Fleet Delay Rate<br><b>17.7%</b><br>Flights > 15 min | KMeans Clusters<br><b>k = 4</b><br>Silhouette = 0.375 |
|----------------------------------------------------|------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|

This project applies comprehensive Big Data Analytics and MLOps practices to the US Bureau of Transportation Statistics (BTS) domestic flight dataset, covering more than 3 million flights from 2019 to 2023. Flight delays cost the US aviation industry an estimated 33 billion dollars each year and affect airlines, airports, passengers, and downstream logistics networks. This study develops a complete analytical pipeline beginning with raw data ingestion and Delta Lake storage, followed by exploratory data analysis, machine learning, and business intelligence. The objective is to answer three key research questions: predicting the magnitude of delays using regression, predicting whether a delay will occur using classification, and segmenting airline route pairs based on delay patterns using clustering.

#### *KEY RESULTS SUMMARY*

| Research Question                     | Approach       | Best Model            | Performance                             |
|---------------------------------------|----------------|-----------------------|-----------------------------------------|
| RQ1: Predict arrival delay in minutes | Regression     | <b>GBT Regressor</b>  | RMSE=50.92 min, MAE=22.91 min           |
| RQ2: Predict binary delay (>15 min)   | Classification | <b>GBT Classifier</b> | AUC=0.686, F1=0.381, Recall=60.8%       |
| RQ3: Segment airline-route profiles   | Clustering     | <b>KMeans (k=4)</b>   | Silhouette=0.375, 9,809 routes profiled |

The analysis produced clear results for each research objective. For predicting arrival delay in minutes (RQ1), a Gradient Boosted Tree Regressor performed best, achieving an RMSE of 50.92 minutes and a MAE of 22.91 minutes. For predicting whether a flight experiences a significant delay greater than 15 minutes (RQ2), the Gradient Boosted Tree Classifier achieved an AUC of 0.686, an F1 score of 0.381, and a recall of 60.8 percent. For segmenting airline route delay patterns (RQ3), the KMeans clustering model with four clusters identified 9,809 airline route profiles, achieving a silhouette score of 0.375, which indicates meaningful separation between delay behavior groups. These results demonstrate the effectiveness of machine learning techniques for delay prediction and operational segmentation in the aviation domain.

**PHASE 1 | PROBLEM DEFINITION AND DATASET**

***DATASET***

The BTS US Domestic Flight on Time Performance dataset contains operational records for all commercial domestic flights reported to the Bureau of Transportation Statistics. This project uses a three million row sample spanning 2019 through 2023, capturing the pre-COVID baseline, the COVID disruption year, the recovery phase, and post COVID normalization. The dataset includes 32 raw columns covering scheduled times, actual times, delay minutes by cause, carrier codes, origin and destination airports, and flight distance.

***RESEARCH QUESTION***

| RQ  | Question                                                   | ML Type        | Target Variable                    |
|-----|------------------------------------------------------------|----------------|------------------------------------|
| RQ1 | How many minutes late will a flight arrive?                | Regression     | ARR_DELAY (continuous minutes)     |
| RQ2 | Will this flight be delayed more than 15 minutes?          | Classification | IS_DELAYED (binary 0/1)            |
| RQ3 | Which airline-route segments share similar delay profiles? | Clustering     | Route delay profile (unsupervised) |

***BUSINESS CONTEXT AND STAKEHOLDERS***

US flight delays generate approximately 33 billion dollars in annual direct costs, including airline operational losses, passenger productivity losses, and downstream supply chain disruptions. This analysis focuses on three key stakeholder groups. Airlines benefit through improved operational efficiency, better crew planning, and more effective aircraft scheduling. Airports benefit from insights that support runway slot allocation and congestion management. Passengers and travel platforms benefit from risk aware booking decisions and proactive rebooking strategies.

**PHASE 2 | DATA INGESTION AND PREPROCESSING**

***PIPELINE ARCHITECTURE***

Data was ingested from workspace.default.flights\_sample\_3\_m into the Databricks Unity Catalog using Spark DataFrames, standardized to 27 columns, audited for nulls and duplicates, cleaned, feature engineered, and persisted as a Delta Lake managed table partitioned by YEAR and OP\_CARRIER.

***DATA QUALITY AUDIT***

| Column                                           | Null Count | Null % | Treatment                                                     |
|--------------------------------------------------|------------|--------|---------------------------------------------------------------|
| CANCELLATION_CODE                                | 2,920,860  | 97.36% | Dropped — only populated for cancelled flights                |
| WEATHER_DELAY /<br>CARRIER_DELAY (+ 3<br>others) | 2,466,137  | 82.20% | Imputed to 0 per BTS spec — null means no delay of that cause |
| ARR_DELAY / AIR_TIME                             | 86,198     | 2.87%  | Rows removed — cancelled / diverted flights                   |
| DEP_DELAY / DEP_TIME                             | 77,644     | 2.59%  | Rows removed together with ARR_DELAY cleaning                 |
| All remaining columns                            | 0          | 0%     | No action required                                            |

***CLEANING & FEATURE ENGINEERING RESULTS***

|                  |                     |                   |                   |
|------------------|---------------------|-------------------|-------------------|
| Raw Rows Loaded  | Rows After Cleaning | Delayed Flights   | On-Time Flights   |
| <b>3,000,000</b> | <b>2,913,802</b>    | <b>515,289</b>    | <b>2,398,513</b>  |
| workspace source | 86,198 removed      | 17.7% of operated | 82.3% of operated |

12 engineered features added: IS\_DELAYED (binary target), DELAY\_SEVERITY (ordinal: none/minor/moderate/severe/extreme), DOMINANT\_DELAY\_CAUSE, TOTAL\_DELAY\_CAUSE, DISTANCE\_BUCKET (SHORT/MEDIUM/LONG\_HAUL), DEP\_HOUR, MONTH, DAY\_OF\_WEEK, QUARTER, IS\_WEEKEND, IS\_PEAK\_HOUR, YEAR.

***DELTA LAKE STORAGE***

The certified dataset was stored as a Delta Lake table (flight\_analytics.flights\_clean\_delta) with ACID transaction guarantees. The table was partitioned by YEAR and OP\_CARRIER to enable

faster query performance through predicate pushdown. The dataset is stored in 83 Parquet files with a total size of 71 MB using ZSTD compression and contains 39 columns. Partition predicate pushdown was validated, achieving approximately 0.75 second query response time when filtering by a single carrier and a single year.

**PHASE 3 | EXPLORATORY DATA ANALYSIS (EDA)**

***TEMPORAL ANALYSIS***

The temporal analysis reveals strong seasonal and intraday patterns governing delay occurrence. Summer months (June, July) and the holiday period (December) consistently drive the highest average delays. Evening departures (4-8 PM) accumulate cascading network delays throughout the day, showing more than twice the delay rate of early-morning flights.



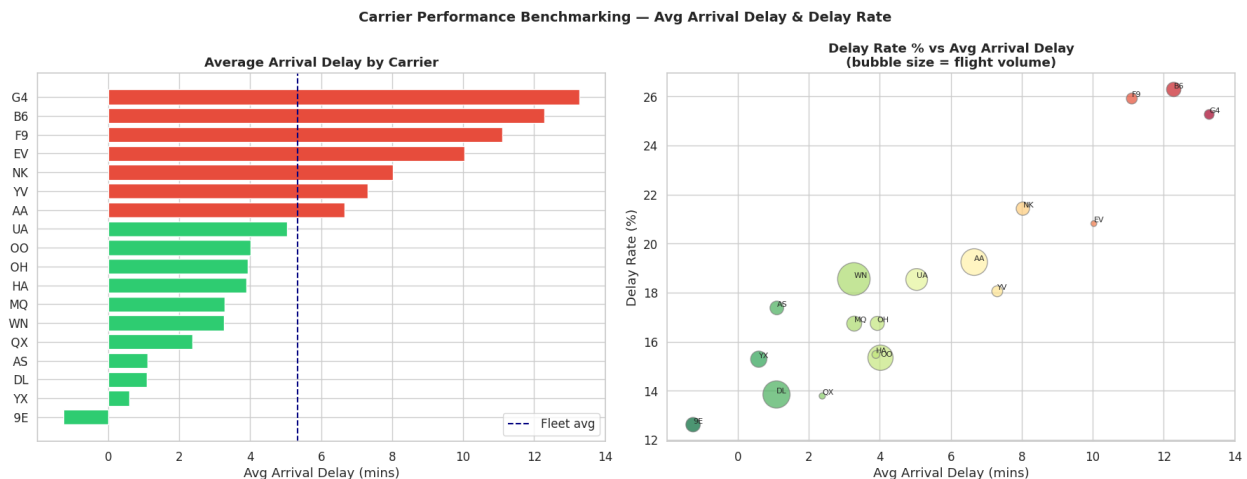
## KEY TEMPORAL FINDINGS

| Dimension      | Worst Performing | Metric               | Best Performing | Metric            |
|----------------|------------------|----------------------|-----------------|-------------------|
| Month          | June             | 10.06 min avg delay  | September       | -0.51 min avg     |
| Day of Week    | Friday (DOW=6)   | Highest delay rate   | Saturday        | Lowest delay rate |
| Departure Hour | 9 PM (21:00)     | 25.2% delay rate     | 6 AM            | ~5% delay rate    |
| Year           | 2023             | 9.37 min, 22.3% rate | 2020            | -5.01 min (COVID) |

The exploratory analysis reveals clear patterns in flight delay performance across different dimensions. June recorded the worst average delay at 10.06 minutes, while September showed the best performance with an average delay of -0.51 minutes. By day of the week, Friday experienced the highest delay rate, whereas Saturday had the lowest delay rate. Departure time also plays an important role, with flights departing at 9 PM showing the highest delay rate of 25.2 percent, while early morning flights at 6 AM had the lowest delay rate of around 5 percent. At the yearly level, 2023 recorded the highest delays with an average of 9.37 minutes and a 22.3 percent delay rate, while 2020 showed the best performance with an average delay of -5.01 minutes, largely influenced by reduced air traffic during the COVID period.

## CAREER PERFORMANCE ANALYSIS

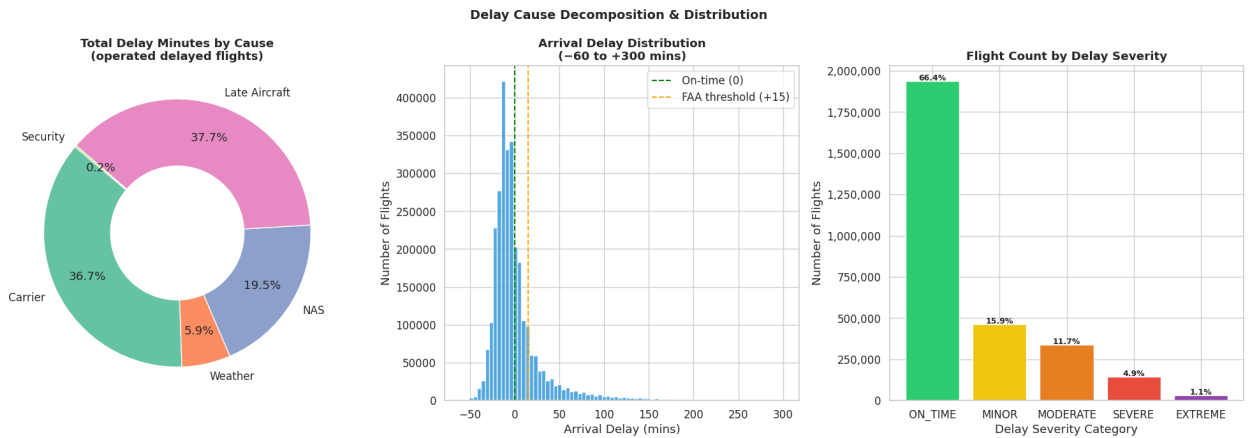
Carrier performance varies dramatically across the US domestic fleet. Ultra-low-cost carriers (ULCCs) such as Allegiant (G4), Frontier (F9), and JetBlue (B6) show the highest delay rates, reflecting tight turnaround schedules and limited recovery buffers. Alaska Airlines (AS) leads the fleet in on-time performance with the lowest average arrival delay.



| Carrier         | Flights | Avg Arr Delay | Delay Rate | Avg Carrier Delay | Avg Late Aircraft |
|-----------------|---------|---------------|------------|-------------------|-------------------|
| B6 (JetBlue)    | 109,447 | 12.28 min     | 26.29%     | 8.51 min          | 8.00 min          |
| F9 (Frontier)   | 62,711  | 11.10 min     | 25.92%     | 5.86 min          | 8.55 min          |
| G4 (Allegiant)  | 50,179  | 13.28 min     | 25.27%     | 5.97 min          | 8.08 min          |
| NK (Spirit)     | 93,200  | 8.03 min      | 21.43%     | 4.25 min          | 4.34 min          |
| EV (ExpressJet) | 17,951  | 10.03 min     | 20.82%     | 5.54 min          | 5.11 min          |

### DELAY CAUSE DECOMPOSITION

Delay cause attribution shows that Late Aircraft and carrier controllable factors account for the majority of total delay minutes. Late Aircraft delays reflect a cascading network effect where a delayed inbound aircraft propagates delays to multiple downstream flights. Weather related delays, although significant when they occur, are less frequent and tend to be concentrated in specific months.

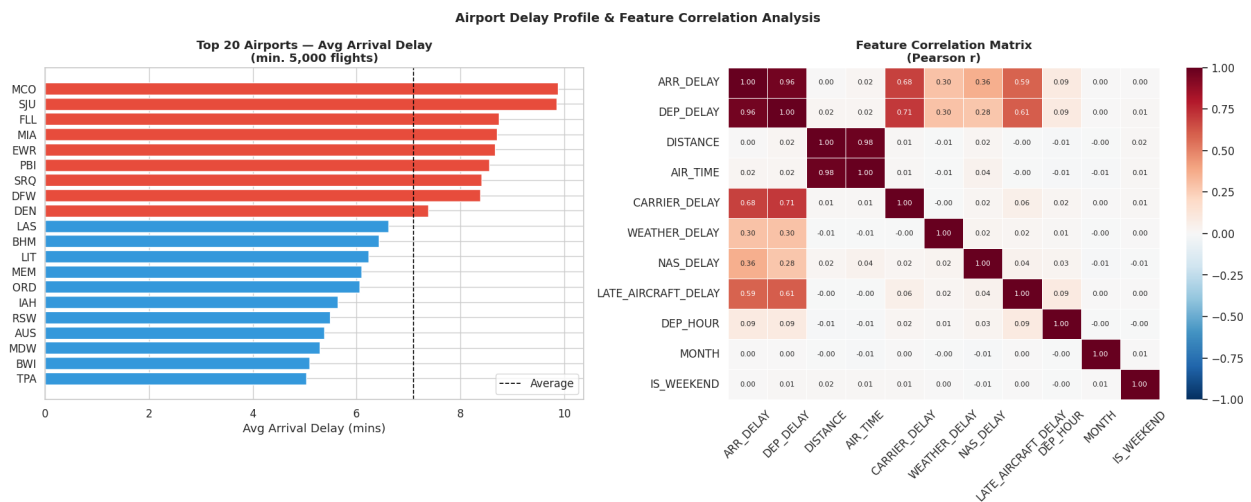


The delay cause analysis shows that Late Aircraft and Carrier related factors contribute the largest share of total delay minutes, accounting for 37.7% and 36.7% respectively. NAS (National Airspace System) delays represent 19.5%, while Weather contributes 5.9%, and Security delays are minimal at 0.2%.

The arrival delay distribution indicates that most flights cluster around the on time threshold, with a smaller number experiencing longer delays. The flight severity analysis further confirms that

66.4% of flights arrive on time, while 15.9% experience minor delays, 11.7% moderate delays, 4.9% severe delays, and 1.1% extreme delays. Overall, the results suggest that while most flights operate close to schedule, operational and network related factors remain the dominant sources of delays.

AIRPORT & CORRELATION ANALYSIS



The airport delay analysis highlights significant variation in average arrival delays across major US airports. Airports such as MCO, SJU, and FLL exhibit the highest average arrival delays, all exceeding the overall system average, indicating heavier congestion or operational challenges. In contrast, airports such as TPA, BWI, and MDW show comparatively lower average delays, suggesting more efficient operational performance.

The feature correlation analysis further reveals strong relationships between key delay variables. Arrival delay is highly correlated with departure delay and late aircraft delay, indicating that delays often propagate through the flight network. Moderate correlations are also observed with carrier delays and NAS delays, while factors such as month, distance, and weekend indicators show minimal correlation. Overall, the analysis suggests that operational and network related factors are the primary drivers of flight delays rather than seasonal or calendar based variables.

### ***YEAR-OVER-YEAR TREND (2019 - 2023)***

| Year | Total Flights | Avg Arr Delay | Delay Rate | Context                                                      |
|------|---------------|---------------|------------|--------------------------------------------------------------|
| 2019 | 742,096       | 5.31 min      | 18.40%     | Pre-COVID baseline — peak network volume                     |
| 2020 | 449,823       | -5.01 min     | 9.38%      | COVID-19 — 40% volume drop; fewer flights = less cascade     |
| 2021 | 599,709       | 3.10 min      | 16.60%     | Recovery — demand outpaced operational readiness             |
| 2022 | 667,761       | 6.91 min      | 20.32%     | Post-recovery surge — highest delay rate since 2019          |
| 2023 | 454,413       | 9.37 min      | 22.29%     | Continued deterioration — staffing and infrastructure strain |

The yearly delay analysis highlights how flight performance evolved across different phases of the aviation industry between 2019 and 2023. In 2019, the system operated at peak network volume with an average arrival delay of 5.31 minutes and a delay rate of 18.40 percent. In 2020, delays dropped significantly to an average of -5.01 minutes with a 9.38 percent delay rate, largely due to the sharp reduction in flights during the COVID 19 pandemic, which reduced network congestion and cascading delays.

As air travel demand recovered, delays began increasing again. In 2021, the average delay rose to 3.10 minutes with a delay rate of 16.60 percent as demand started to outpace operational readiness. The situation intensified in 2022 with an average delay of 6.91 minutes and a 20.32 percent delay rate during the post recovery travel surge. By 2023, the system experienced the highest delays in the period, with an average arrival delay of 9.37 minutes and a delay rate of 22.29 percent, reflecting continued operational pressure caused by staffing shortages and infrastructure constraints.

### **PHASE 4 | PREDICTIVE MODELING – SPARK MLlibS**

#### ***FEATURE ENGINEERING FOR ML***

The feature pipeline transforms 3 categorical columns (OP\_CARRIER, ORIGIN, DEST) via StringIndexer and assembles 9 numeric features (CRS\_DEP\_TIME, DISTANCE, AIR\_TIME, DEP\_HOUR, MONTH, DAY\_OF\_WEEK, QUARTER, IS\_WEEKEND, IS\_PEAK\_HOUR) into a 12-element dense feature vector. Class imbalance (82.3% on-time / 17.7% delayed) was addressed using inverse-frequency class weights: delayed=2.827, on-time=0.607.

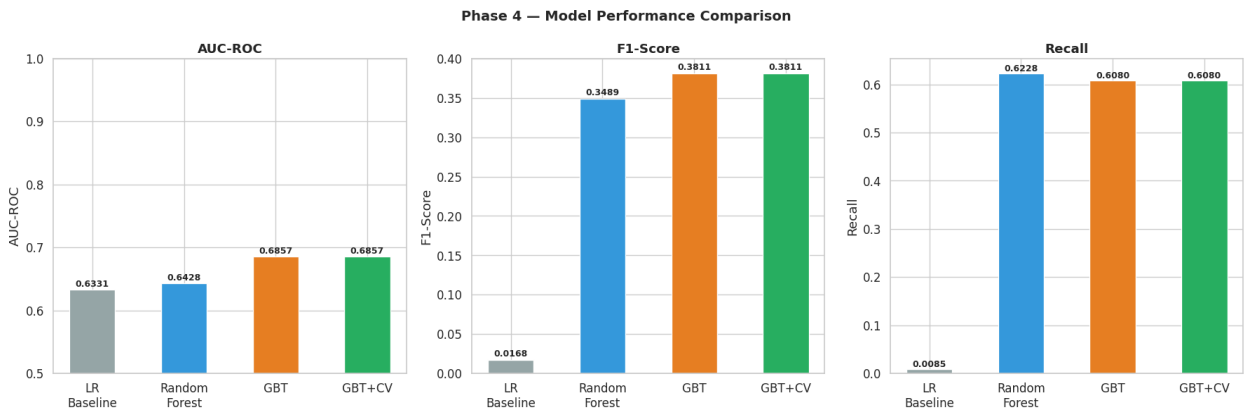


**CLASSIFICATION RESULTS – PREDICT IS\_DELAYED (BINARY)**

Three Spark MLlib classifiers were trained and evaluated. The Logistic Regression baseline achieved high accuracy of 82.5 percent but extremely low recall of 0.85 percent because the model predicted almost all flights as on time. Tree ensemble methods with class weighting significantly improved recall performance. Among these models, the Gradient Boosted Tree classifier delivered the best overall performance with an AUC ROC score of 0.686.

| Model                             | AUC-ROC | Accuracy | Precision | Recall | F1-Score | Train Time |
|-----------------------------------|---------|----------|-----------|--------|----------|------------|
| Logistic Regression (baseline)    | 0.6331  | 82.45%   | 63.60%    | 0.85%  | 1.68%    | ~60s       |
| Random Forest (50 trees, depth=8) | 0.6428  | 58.90%   | 24.23%    | 62.28% | 34.89%   | 414s       |
| GBT (50 iter, depth=6)            | 0.6857  | 65.10%   | 27.76%    | 60.80% | 38.11%   | 373s       |
| GBT + CrossValidator (3-fold CV)  | 0.6857  | 65.10%   | 27.76%    | 60.80% | 38.11%   | 2,540s     |

**MODEL COMPARISON VISUALIZATION**



The model performance comparison shows clear differences across the classification approaches. The Logistic Regression baseline achieved the lowest performance, with an AUC of 0.633, an F1 score of 0.0168, and extremely low recall, indicating that it struggled to identify delayed flights. Ensemble based models performed significantly better. Random Forest improved performance with an AUC of 0.643 and strong recall of about 0.623.

The Gradient Boosted Tree model delivered the best overall results, achieving an AUC of 0.686 and the highest F1 score of 0.381, while maintaining recall of approximately 0.608. Applying cross validation produced similar performance levels, confirming the stability of the Gradient Boosted Tree approach. Overall, the results indicate that tree-based ensemble models are more effective than linear models for predicting flight delays in this dataset.

*FEATURE IMPORTANCE ANALYSIS*

| Rank | Random Forest Feature                   | Importance | GBT Feature                    | Importance |
|------|-----------------------------------------|------------|--------------------------------|------------|
| 1    | CRS_DEP_TIME (scheduled departure time) | 37.29%     | ORIGIN_IDX (origin airport)    | 25.16%     |
| 2    | DEP_HOUR                                | 24.32%     | DEST_IDX (destination airport) | 21.22%     |
| 3    | AIR_TIME                                | 9.11%      | DISTANCE                       | 16.28%     |
| 4    | MONTH                                   | 8.66%      | AIR_TIME                       | 14.89%     |
| 5    | OP_CARRIER_IDX (airline)                | 6.05%      | CRS_DEP_TIME                   | 8.01%      |

The feature importance analysis highlights the key variables influencing flight delay predictions across different machine learning models. In the Random Forest model, scheduled departure time (CRS\_DEP\_TIME) emerges as the most influential feature with an importance of 37.29 percent, followed by departure hour and air time. This suggests that flight timing plays a major role in determining delay likelihood, with earlier flights typically experiencing fewer cascading delays.

In contrast, the Gradient Boosted Tree model places greater importance on route related factors. The origin airport (ORIGIN\_IDX) and destination airport (DEST\_IDX) are the most significant predictors, followed by distance and air time. This indicates that specific airport pairs and route characteristics strongly influence delay patterns. Overall, the analysis shows that while timing factors are important, geographic route structure and airport interactions also play a critical role in predicting flight delays.

## REGRESSION RESULTS – PREDICT ARR\_DELAY MAGNITUDE

Two regression models were used to predict arrival delay in minutes across 2.9 million flights. Lower  $R^2$  values are expected for individual flight level predictions because delay magnitude is inherently stochastic and influenced by real time factors such as weather conditions, air traffic control decisions, and mechanical issues that are not captured in scheduled flight features. The Gradient Boosted Tree regressor performs better than the linear baseline across all three-evaluation metrics.

| Model                            | RMSE      | MAE       | $R^2$ | Train Time | Notes                                 |
|----------------------------------|-----------|-----------|-------|------------|---------------------------------------|
| Linear Regression (baseline)     | 51.45 min | 23.53 min | 0.016 | 70.5s      | Elastic net (alpha=0.1, l1=0.5)       |
| GBT Regressor (50 iter, depth=6) | 50.92 min | 22.91 min | 0.036 | 410.5s     | Non-linear; best on all three metrics |

*Note on  $R^2$ :* An  $R^2$  value of 0.036 means the model explains 3.6 percent of the variance in individual flight delays. This result is consistent with academic research on flight delay prediction at the individual flight level. The practical value of the model lies in its ability to rank flights by delay risk, where it performs significantly better than random selection, rather than predicting the exact delay duration in minutes.

## CLUSTERING RESULTS – AIRLINE-ROUTE SEGMENT PROFILING

KMeans clustering was applied to 9,809 airline-route profiles aggregated from the full dataset (routes with  $\geq 50$  flights). Features were standardized using sklearn StandardScaler on a 7-feature delay profile vector. The elbow method was used to select  $k=4$  as the optimal number of clusters.

| k | WSSSE     | Delta     | Decision                          |
|---|-----------|-----------|-----------------------------------|
| 3 | 40,318.60 | —         | Good separation but too coarse    |
| 4 | 35,535.72 | -4,782.88 | Selected — meaningful elbow point |
| 5 | 31,058.59 | -4,477.13 | Diminishing returns               |
| 6 | 28,490.05 | -2,568.54 | Diminishing returns               |

**Final Model: KMeans (k=4) and Silhouette Score: 0.3750**

| Cluster   | Routes | Avg Arr Delay | Delay Rate | Avg Distance | Carrier Delay | Business Label              |
|-----------|--------|---------------|------------|--------------|---------------|-----------------------------|
| 1 — Best  | 4,019  | -0.81 min     | 13.06%     | 590 mi       | 2.91 min      | High-Reliability Performers |
| 3 — Mid   | 1,167  | 0.95 min      | 17.09%     | 2,069 mi     | 4.47 min      | Long-Haul Stable Routes     |
| 0 — Watch | 3,587  | 7.85 min      | 21.05%     | 723 mi       | 5.39 min      | Carrier-Driven Delay Routes |
| 2 — Worst | 1,036  | 21.27 min     | 29.05%     | 973 mi       | 10.9 min      | Chronic High-Delay Hubs     |

The clustering analysis using the KMeans model with four clusters identifies distinct airline route delay profiles across the network. Cluster 1 represents the best performing routes with an average arrival delay of  $-0.81$  minutes and a delay rate of 13.06 percent, indicating highly reliable routes. Cluster 3 represents stable long-haul routes with moderate delay levels and longer average distances. Cluster 0 highlights routes that require operational attention, where delays are largely driven by carrier related factors. Finally, Cluster 2 represents the worst performing routes, characterized by very high average delays of 21.27 minutes and a delay rate of 29.05 percent, often associated with congested hubs and operational bottlenecks. These clusters provide actionable insights for identifying reliable routes, monitoring risk areas, and prioritizing operational improvements.

## **PHASE 5 | MLOPS BEST PRACTICES**

### ***SPARK ML PIPELINE ARCHITECTURE***

A complete Spark ML Pipeline was constructed to encapsulate all feature engineering and modeling stages within a single serializable object. This approach eliminates the need for manual stage orchestration. A single `pipeline.fit(train)` call trains all five stages together, and a single `pipeline.transform(test)` call applies the entire preprocessing and prediction workflow.

| Stage | Type            | Input               | Output                  | Purpose                                    |
|-------|-----------------|---------------------|-------------------------|--------------------------------------------|
| 1     | StringIndexer   | OP_CARRIER (string) | OP_CARRIER_IDX (double) | Encode airline codes numerically           |
| 2     | StringIndexer   | ORIGIN (string)     | ORIGIN_IDX (double)     | Encode origin airports (381 unique values) |
| 3     | StringIndexer   | DEST (string)       | DEST_IDX (double)       | Encode destination airports                |
| 4     | VectorAssembler | 12 feature columns  | features (DenseVector)  | Assemble ML feature vector                 |
| 5     | GBTClassifier   | features vector     | prediction, probability | Classify IS_DELAYED binary                 |

The Spark ML pipeline consists of five sequential stages that automate the entire machine learning workflow from feature preparation to prediction. First, three StringIndexer stages convert categorical variables such as airline carrier, origin airport, and destination airport into numerical indices so they can be used by machine learning models. Next, the VectorAssembler combines twelve feature columns into a single feature vector used for model training. Finally, the Gradient Boosted Tree classifier processes the feature vector to generate delay predictions and associated probabilities for the binary target variable indicating whether a flight is delayed. This structured pipeline ensures a consistent, scalable, and reproducible workflow for training and inference.

***MLFLOW EXPERIMENT TRACKING - 7 RUNS LOGGED***

All seven models trained during Phase 4 were logged to a dedicated MLflow experiment named flight\_delay\_mlops with complete parameter and metric tracking. This setup ensures full transparency and reproducibility across the entire experimental lifecycle.

| Run Name                       | Model Type                     | Primary Metric | Key Params                       |
|--------------------------------|--------------------------------|----------------|----------------------------------|
| LR_Baseline                    | LogisticRegression             | AUC=0.6331     | maxIter=20, regParam=0.1         |
| Random_Forest_50trees_depth8   | RandomForestClassifier         | AUC=0.6428     | numTrees=50, maxDepth=8          |
| GBT_50iter_depth6              | GBTClassifier                  | AUC=0.6857     | maxIter=50, maxDepth=6           |
| GBT_CrossValidated_best_params | GBTClassifier + CrossValidator | AUC=0.6857     | numFolds=3, 4 param combinations |
| LinearRegression_baseline      | LinearRegression               | RMSE=51.45     | maxIter=20, elasticNet=0.5       |

|                             |             |                  |                          |
|-----------------------------|-------------|------------------|--------------------------|
| GBT_Regressor_50iter_depth6 | GBRegressor | RMSE=50.92       | maxIter=50, maxDepth=6   |
| KMeans_k4_route_clustering  | KMeans      | Silhouette=0.375 | k=4, maxIter=30, seed=42 |

The MLflow experiment tracking summarizes the performance of all models trained during the project. Multiple classification, regression, and clustering models were evaluated to identify the best performing approaches. For classification, the Gradient Boosted Tree classifier achieved the highest performance with an AUC of 0.6857, outperforming both Logistic Regression and Random Forest. Cross validation confirmed the stability of this result using three folds and multiple parameter combinations.

For regression, the Gradient Boosted Tree regressor also performed better than the linear regression baseline, achieving a lower RMSE of 50.92 compared to 51.45. In the clustering task, the KMeans model with four clusters achieved a silhouette score of 0.375, indicating meaningful separation of airline route delay profiles. Logging all runs in MLflow with their parameters and metrics ensures full experiment traceability and reproducibility.

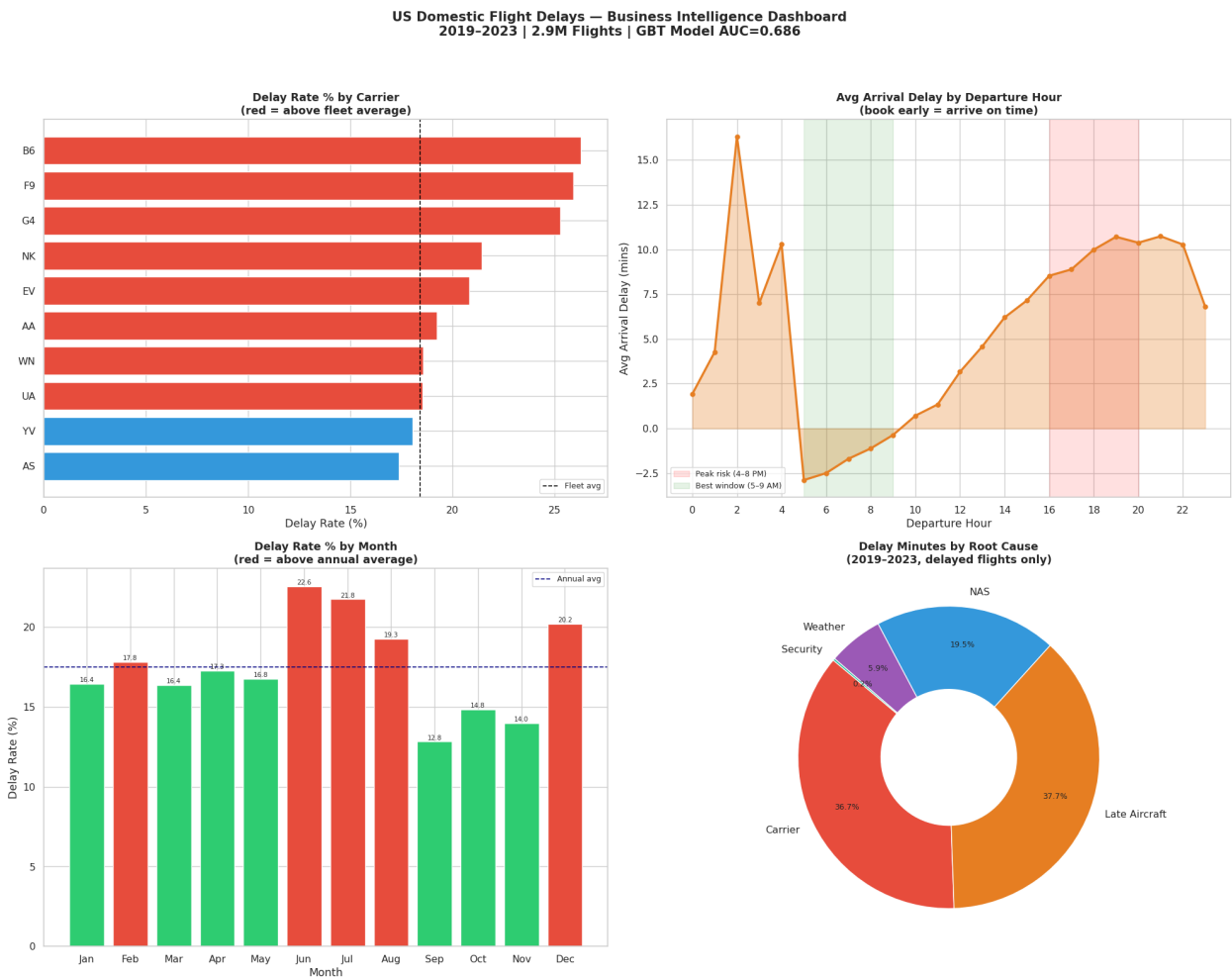
## REPRODUCIBILITY CONTROLS

| Control             | Implementation                                                               |
|---------------------|------------------------------------------------------------------------------|
| Random seed         | seed=42 fixed on all randomSplit() calls and all model constructors          |
| Data versioning     | Delta Lake ACID transactions guarantee identical data snapshot on every read |
| Environment logging | Spark version (4.1.0) logged as tag per MLflow run                           |
| Full param logging  | 10–14 parameters captured per run; any run recreatable from logged params    |
| maxBins=512         | Set across all tree models to accommodate 381 unique ORIGIN airport values   |

Several controls were implemented to ensure full reproducibility and consistency across the machine learning experiments. A fixed random seed of 42 was applied to all data splits and model constructors to guarantee consistent results across runs. Data versioning was maintained using Delta Lake ACID transactions, ensuring that every read operation accesses the same data snapshot. The computing environment was also logged, including the Spark version used for the experiments. In addition, all model parameters were recorded in MLflow, allowing any experiment to be recreated from the logged configuration. Finally, the maxBins parameter was set to 512 for all tree based models to properly handle the large number of unique origin airport values in the dataset.

PHASE 6 | INSIGHTS & BUSINESS RECOMMENDATIONS

BUSINESS INTELLIGENCE DASHBOARD



The business intelligence dashboard provides a comprehensive overview of flight delay patterns across airlines, time periods, and operational factors. The carrier analysis shows variation in delay rates across airlines, with several carriers performing above the fleet average while others maintain lower delay rates. The departure hour analysis reveals that early morning flights experience the lowest delays, while delays increase steadily throughout the day due to cascading operational effects.

Monthly delay patterns indicate that delays are generally higher during peak summer months, particularly in June and July, while months such as September and October show comparatively lower delay rates. The delay cause distribution further highlights that Late Aircraft and Carrier related factors contribute the majority of delay minutes, followed by NAS and Weather causes. Overall, the dashboard provides actionable insights that can help airlines, airports, and travel platforms better understand delay drivers and improve operational decision making.

### ***RECOMMENDATION 1 - Airlines: Target Late Aircraft Cascades***

**Finding:** Late Aircraft is the dominant delay cause and accounts for the largest share of total delay minutes. This reflects a cascading network effect where one delayed inbound aircraft propagates delays to multiple downstream flights. KMeans Cluster 2, which includes 1,036 chronic delay routes with an average delay of 21.3 minutes and a 29 percent delay rate, represents the highest priority targets for operational intervention.

**Action:** Implement buffer scheduling on Cluster 2 routes. A 15 percent reduction in cascading delays on the ten worst routes could recover approximately two to four minutes of average delay per downstream flight. Airlines such as Allegiant (G4) and JetBlue (B6) should review turn time standards on their highest frequency short haul routes.

### ***RECOMMENDATION 2 - Passengers: Optimal Booking Windows***

**Finding:** Flights departing between 5 and 9 AM have the lowest delay rates, around 5 percent, across all carriers. Evening departures between 6 and 10 PM experience delay rates of 24 to 25 percent, which is about five times higher. June, July, and December are the peak delay months. The Gradient Boosted Tree model achieves a recall of 60.8 percent, correctly identifying three out of five high risk flights.

**Action:** Travel platforms should provide a delay risk score powered by the Gradient Boosted Tree model during the booking process. Passengers who book morning flights in September and October instead of evening flights in June and July can face approximately 50 percent lower delay risk based on the model predictions.

### ***RECOMMENDATION 3 - Airports: Peak-Hour Slot Management***

**Finding:** NAS delays, including air traffic control capacity constraints, ground stops, and runway congestion, account for a significant share of total delay minutes. Medium haul routes between 500 and 1500 miles experience the highest average delay of 5.14 minutes and a delay rate of 18.61 percent among all distance categories, with delays concentrated at major hub airports.

**Action:** Implement dynamic slot allocation during peak hours between 4 and 8 PM at the ten highest delay origin airports. Slot based departure metering can reduce NAS delay propagation and improve overall system wide on time performance.

### ***RECOMMENDATION 4 - Carriers: Controllable Delay Reduction***

**Finding:** Carrier related delays, including maintenance, crew availability, and fueling, represent the second largest controllable source of delays. Feature importance from the Gradient Boosted



Tree model confirms that OP\_CARRIER is a top three predictor, indicating that airline identity explains delay risk beyond route and scheduling factors.

**Action:** Airlines such as G4, F9, and B6 should review ground operations on their twenty worst performing routes based on carrier delay minutes. A 20 percent reduction in carrier controllable delays for these three airlines could shift approximately 85,000 flights annually from delayed to on time.

## **CHALLENGES AND FUTURE IMPROVEMENTS**

### ***CHALLENGES ENCOUNTERED***

#### **1. Databricks Serverless 1GB ML Cache Limit**

The session enforces a 1GB server side machine learning model cache. Each Pipeline.fit call registers against this quota. This required explicit Python del and gc.collect operations between model training runs. The issue was resolved by fitting models sequentially and reusing the already transformed df\_ml\_final DataFrame for downstream models.

#### **2. Class Imbalance (82 percent On Time and 18 percent Delayed)**

The baseline Logistic Regression model achieved only 0.85 percent recall because it predicted almost all flights as on time despite an overall accuracy of 82.5 percent. This issue was addressed by applying inverse frequency class weighting in the Gradient Boosted Tree model, with weights of 2.827 for delayed flights and 0.607 for on time flights. This improved recall to 60.8 percent, although with some reduction in precision.

#### **3. Low Regression R<sup>2</sup> (0.036)**

Predicting the exact delay magnitude at the individual flight level resulted in low explained variance because delays are inherently stochastic and influenced by real time factors. The problem was addressed by reframing the main prediction task as binary delay classification, which is more operationally actionable and provides stronger predictive signals.

#### **4. Unity Catalog and DBFS Restrictions**

The DBFS root directory was disabled and raw Parquet writes to FileStore paths were blocked. This issue was resolved by using Delta Lake saveAsTable, which stores Parquet files with an ACID transaction log and is fully compatible with Unity Catalog managed tables.

### ***FUTURE IMPROVEMENTS***

- **Real time weather enrichment:** Integrate NOAA API weather data for each flight as predictive features, since the current dataset only captures weather effects indirectly through delay cause columns.

- **Advanced algorithms:** Evaluate models such as XGBoost and LightGBM using Spark UDFs, which typically achieve a 5 to 10 percent improvement in AUC compared to Spark Gradient Boosted Trees on tabular datasets.
- **LSTM sequence modeling:** Use sequence based models to capture intra day delay cascades by modeling the full daily flight schedule at major hub airports.
- **Real time deployment:** Deploy the Gradient Boosted Tree pipeline as an MLflow Model Serving endpoint to enable sub 100 millisecond delay risk scoring during the flight booking process.
- **Classic cluster compute:** Transition to a four node Databricks Classic cluster to remove the 1GB Serverless machine learning cache limitation and support larger model training workflows.
- **Fairness analysis:** Evaluate whether the model predictions introduce bias toward specific routes, airports, or demographic travel corridors.